

Migration Setting

Source Repository (Training Project)

my_project/

train.py

model.py

data.py

utils.py

config.yaml

...

train.py

```
def train(model, loader):  
    model.train()  
    for x, y in loader:  
        optim.zero_grad()  
        loss = model(x, y)  
        loss.backward()  
        optim.step()
```

Migrate to New Ecosystem

Framework B

Framework C

Language X

Initial Migrated Repository (Imperfect)

migrated_project/

train.py

model.py

data.py

utils.py

config.yaml

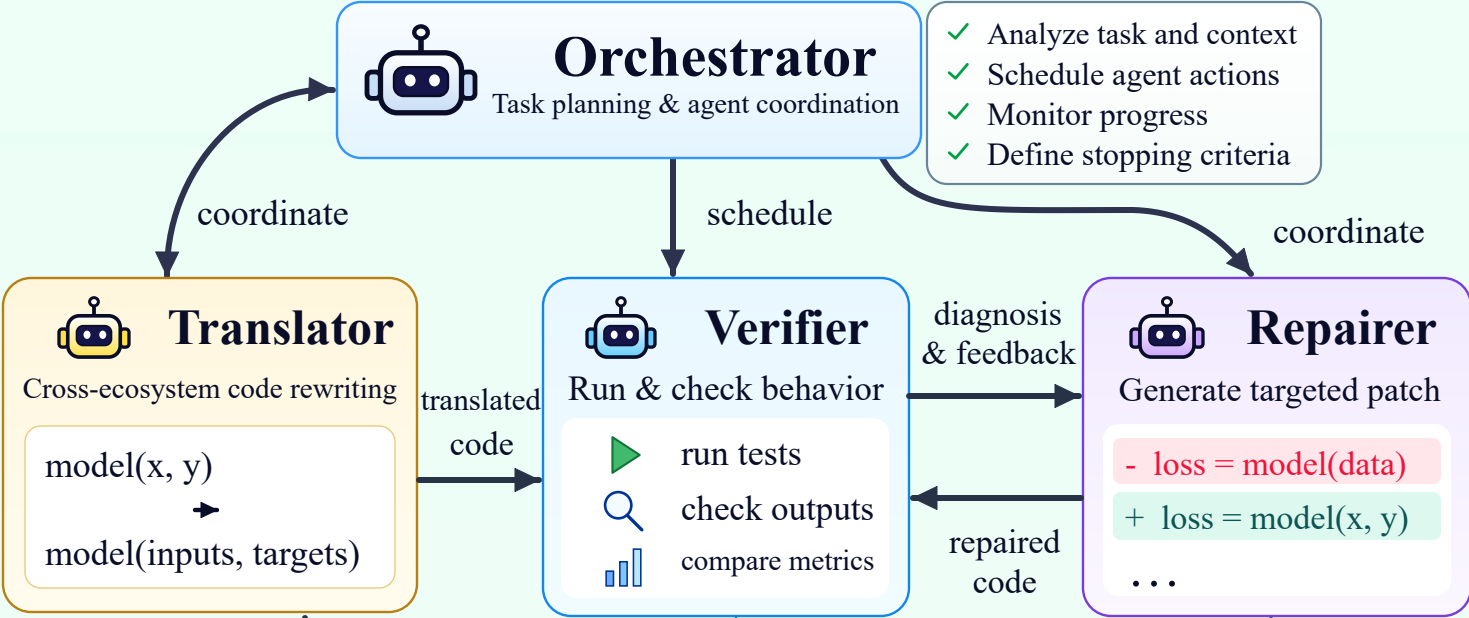
...

train.py

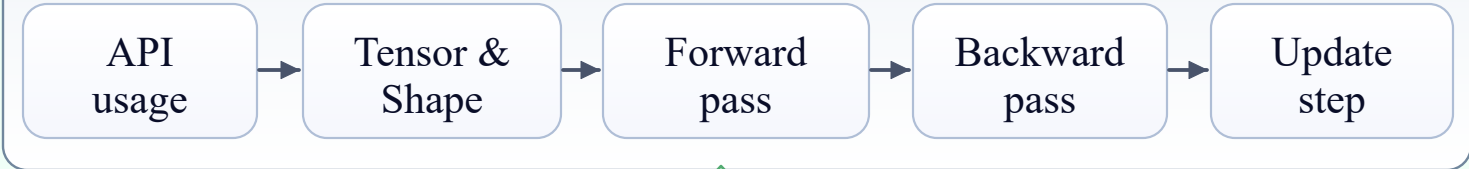
```
def train(model, loader):  
    model.train()  
    loss = model(data)  
    # missing labels?  
    loss.backward()  
    optim.step()  
    ...
```

Initial migration has errors
(may fail to run or produce incorrect behavior)

Dependency Guided Multi-Agent Framework



Training Semantics Diagnosis Chain



Dependency-Aware Workspace

Repository Graph

src/

model.py

train.py

data.py

Evidence Store

Code snippets

Execution traces

Error messages

Context for Next Iteration

Related files

Historical patches

Runtime states

Preserve and update across iterations

Generalization and Efficiency



Cross-framework



Cross-language



Repository-scale



Token-efficient

Verified Output

Repaired Repository

migrated_project/

train.py

model.py

data.py

utils.py

config.yaml

...

train.py

model.py

data.py

utils.py

config.yaml

...

✓

✓

✓

✓

✓

✓

Migration successful
Functionally correct and behaviorally aligned

Verified execution

- ✓ passes (no runtime errors)
- ✓ aligned behavior
- ✓ stable update